

Introduction to Metagenomics

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Why are you here?

- Metagenomics is a trendy, emerging technique
- You feel (in your gut) that you need metagenomics in the future
- You did traditional microbiology / genomics / environmental science and would like to go to a new level
- You want to characterise a specific microbial community
- You know how to generate the data but not how to analyse it

What will you learn on this course?

- Different metagenomics approaches
 - General overview
 - Advantages, disadvantages, **biases**
 - Requirements (knowledge, time, money, hardware)
- Bioinformatics software and pipelines in metagenomics
 - The method behind the software
 - Basic usage examples
 - Specific software names and download links

After this course you will be able to...

- Start planning your own metagenomics project
- Decide on the most appropriate metagenomics approach
- Pay particular attention on the caveats of different methods
- Realise that metagenomics is the way forward in your project
- Realise that metagenomics is not ready yet to answer your question
- Follow the developments on the field

We won't leave you hanging...

- Please feel free to contact us if you need help (Slack)!
- Build up your own network of professionals, help each other
- Learn new methods, approaches by using your new basic knowledge

The tools we will use



What is metagenomics?

- Using genomics / bioinformatics tools to analyse a community of small “things”
- Understanding the microbiome, virome, resistome, etc.
- Who is there and what are they doing?



How did metagenomics grow up?

- We thought we couldn't culture everything
 - Later we realised we would be able to...
- We realised microbes are everywhere (even in some places where they shouldn't be)
 - But then we learned it might have been our own fault
- We know they are there, because we see them (through a microscope), and can get their DNA
- We have suitable tools to sequence / analyse growing amounts of very diverse DNA (NGS)
- We know how important microbial communities are
- We are realising how biased our microbiology knowledge is

Most common metagenomics project questions

- Describe the composition of a microbial community
- Describe the variance within the composition
- Observe significant difference in the composition after...
 - Physiological change
 - Pathophysiological change
 - Environmental change
- Find marker species for a certain microbial community
- Map the interactions within the community or with its surroundings

Specific project questions

- Monitoring and tracking a known pathogen in mixed microbial communities
- Finding and assembling the genome of an unknown species in a low diversity metagenomics sample
- Tracking hygiene and contamination
- ...



Step 1: Take a sample

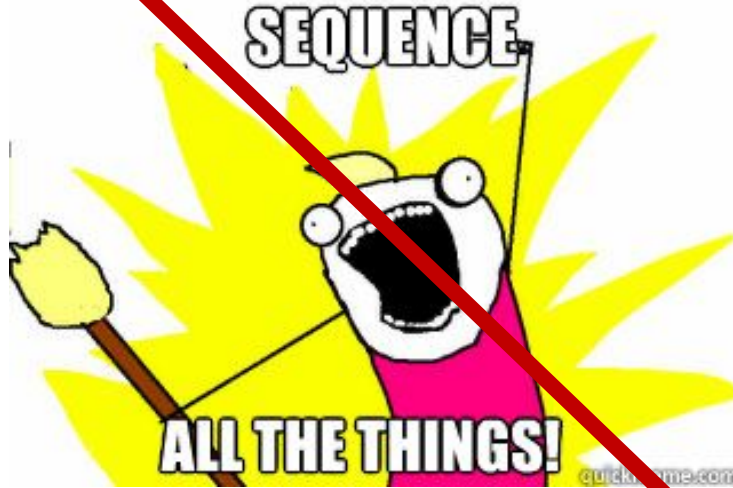


Step ± 2 : Take a sample

Step 0: Define your question

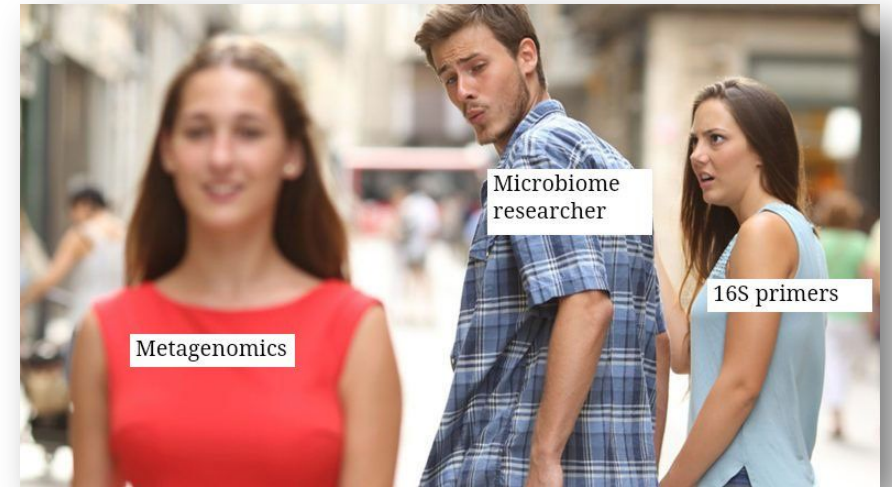
Step 1: Choose an appropriate
approach

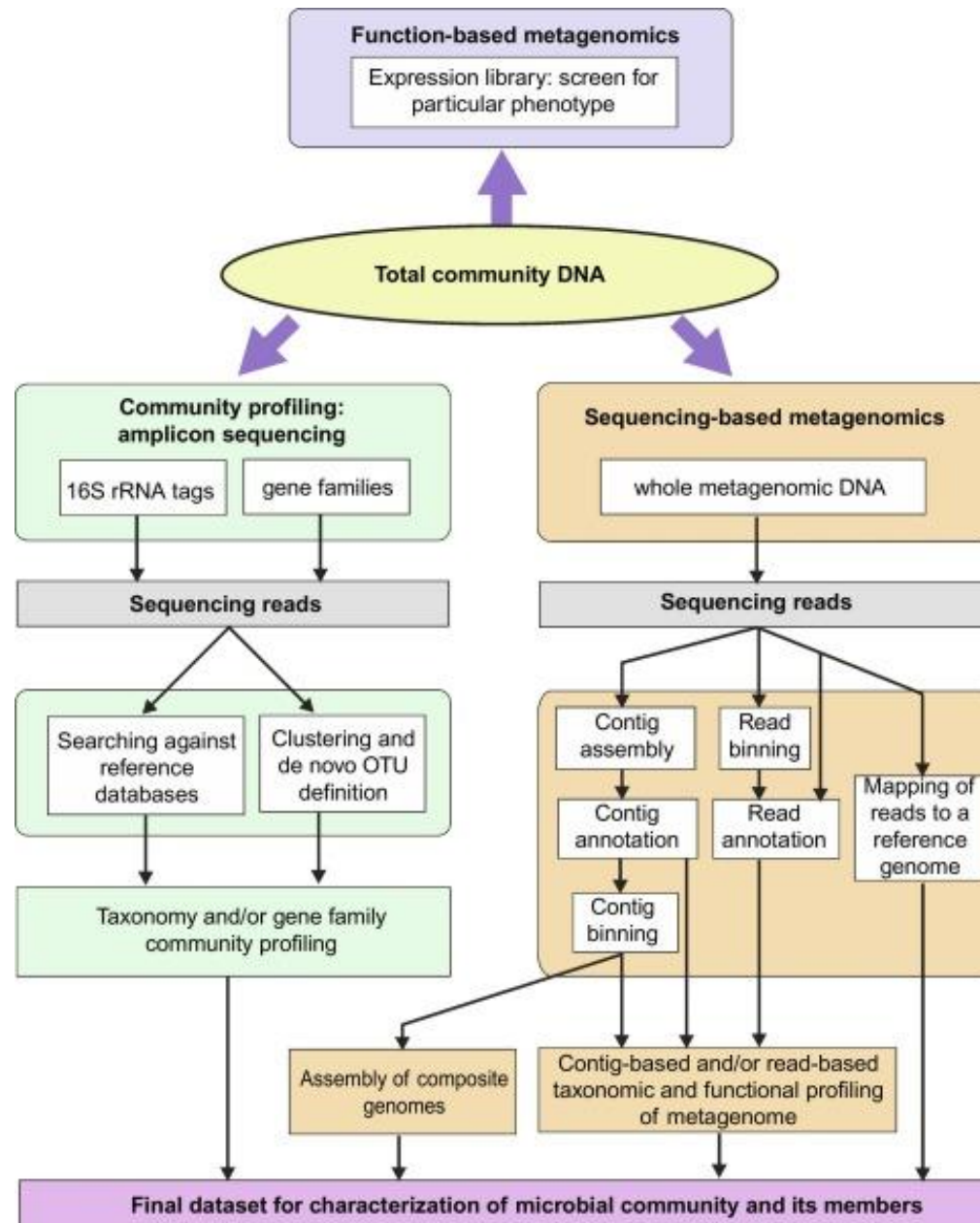
Step 0 – Define your question



Step 1 – Choosing the approach

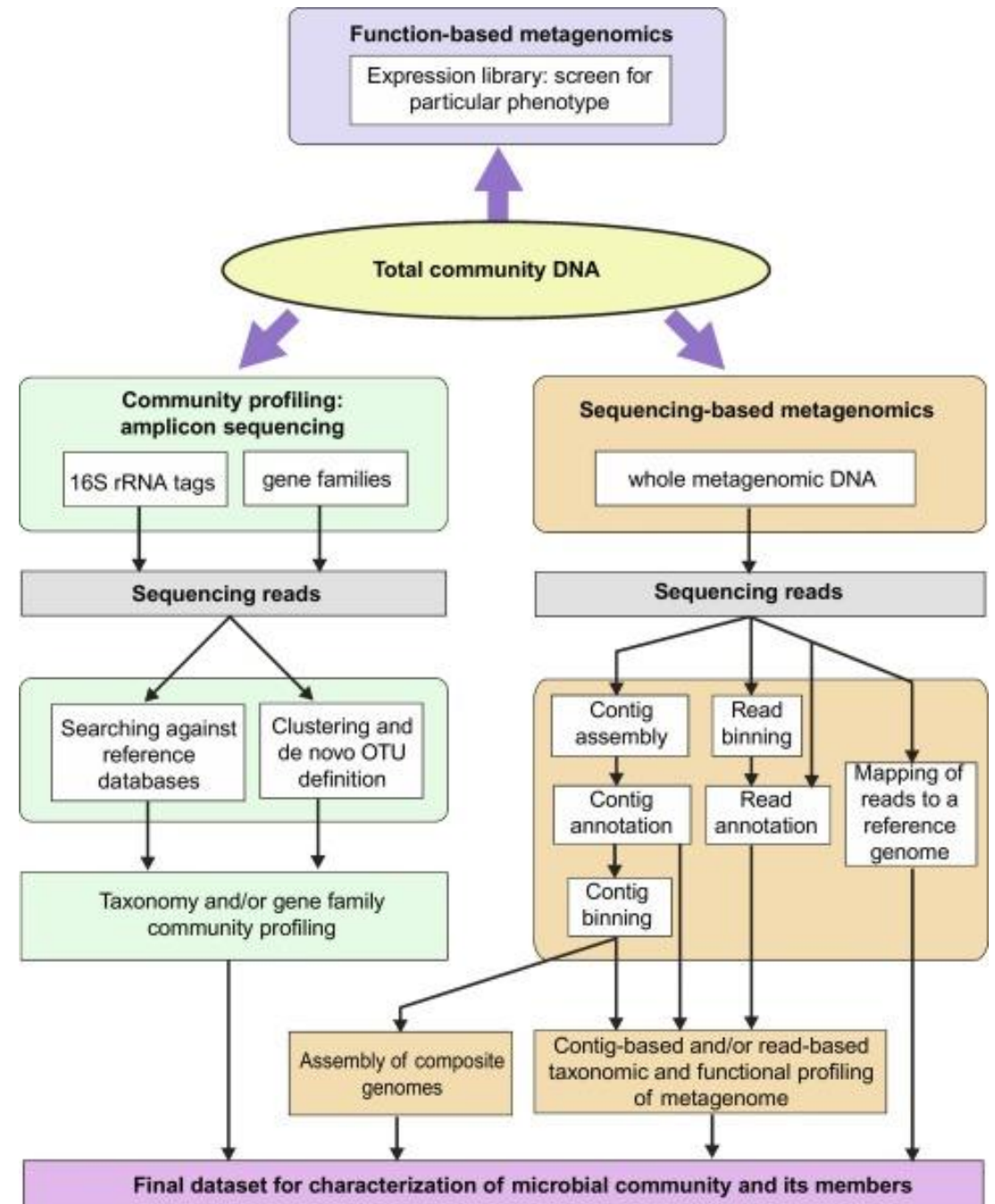
- Define your needs
 - Level of precision / resolution / sensitivity / **reliability**
 - Amount of data
 - Type of answers
- Know your limits
 - Tools / databases
 - Time
 - Lab infrastructure
 - Available hardware
 - Bioinformatics knowledge
 - Money

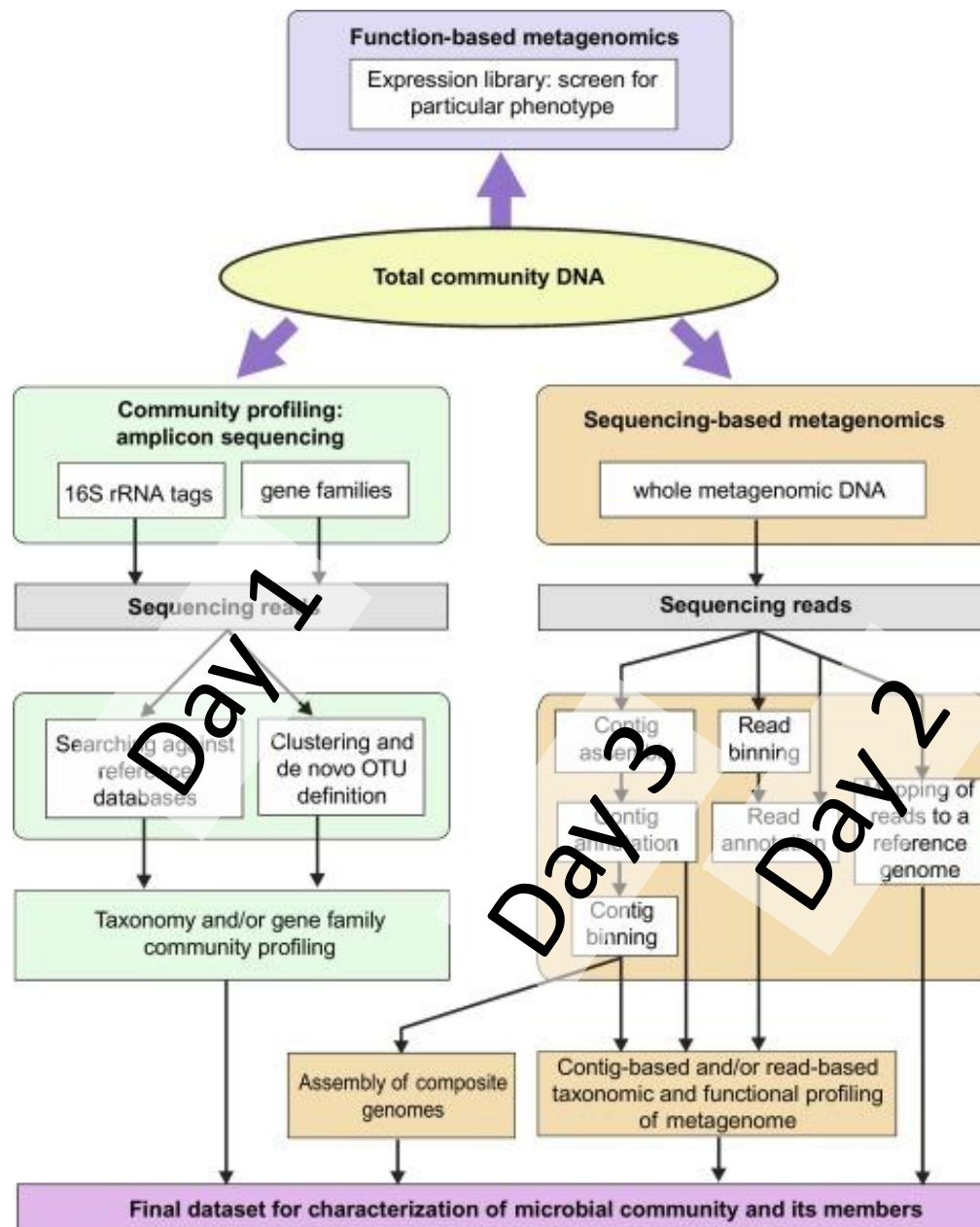




Choosing the approach

- Define your needs
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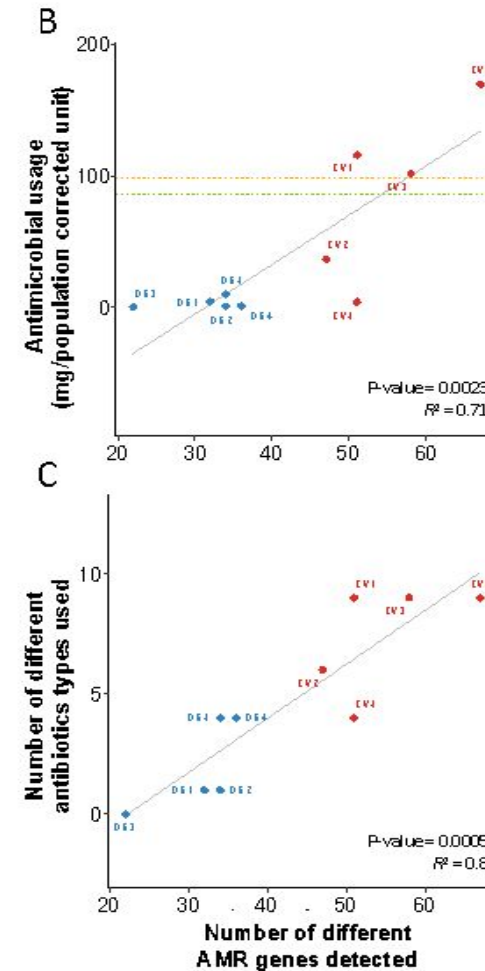
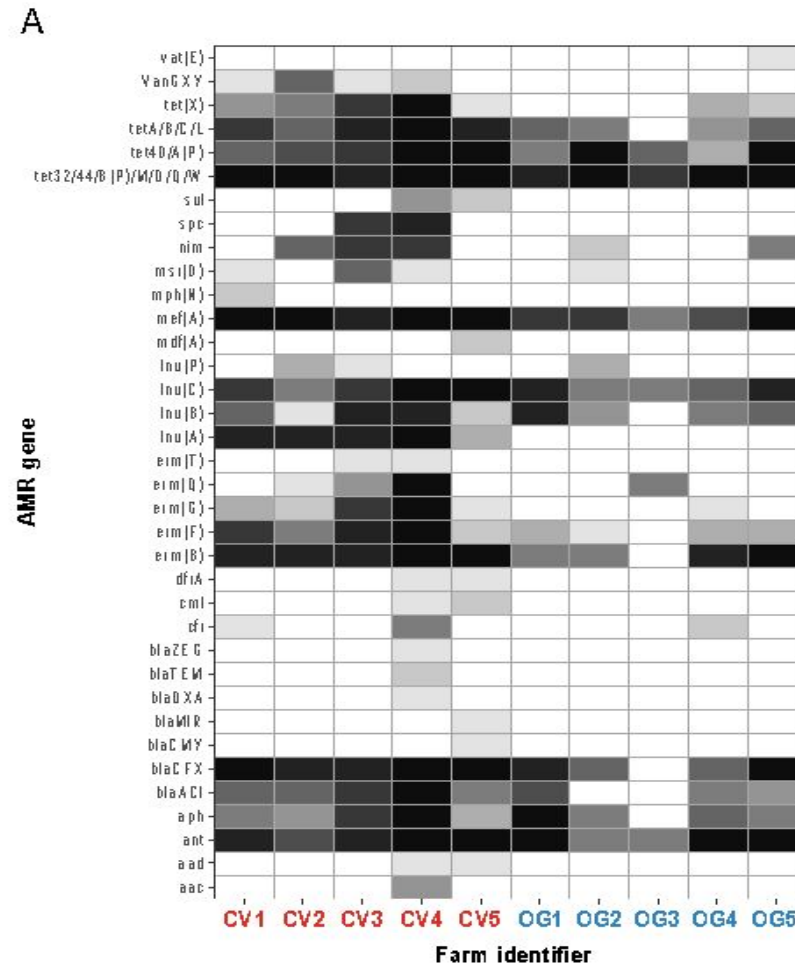


Step 2: Take a sample

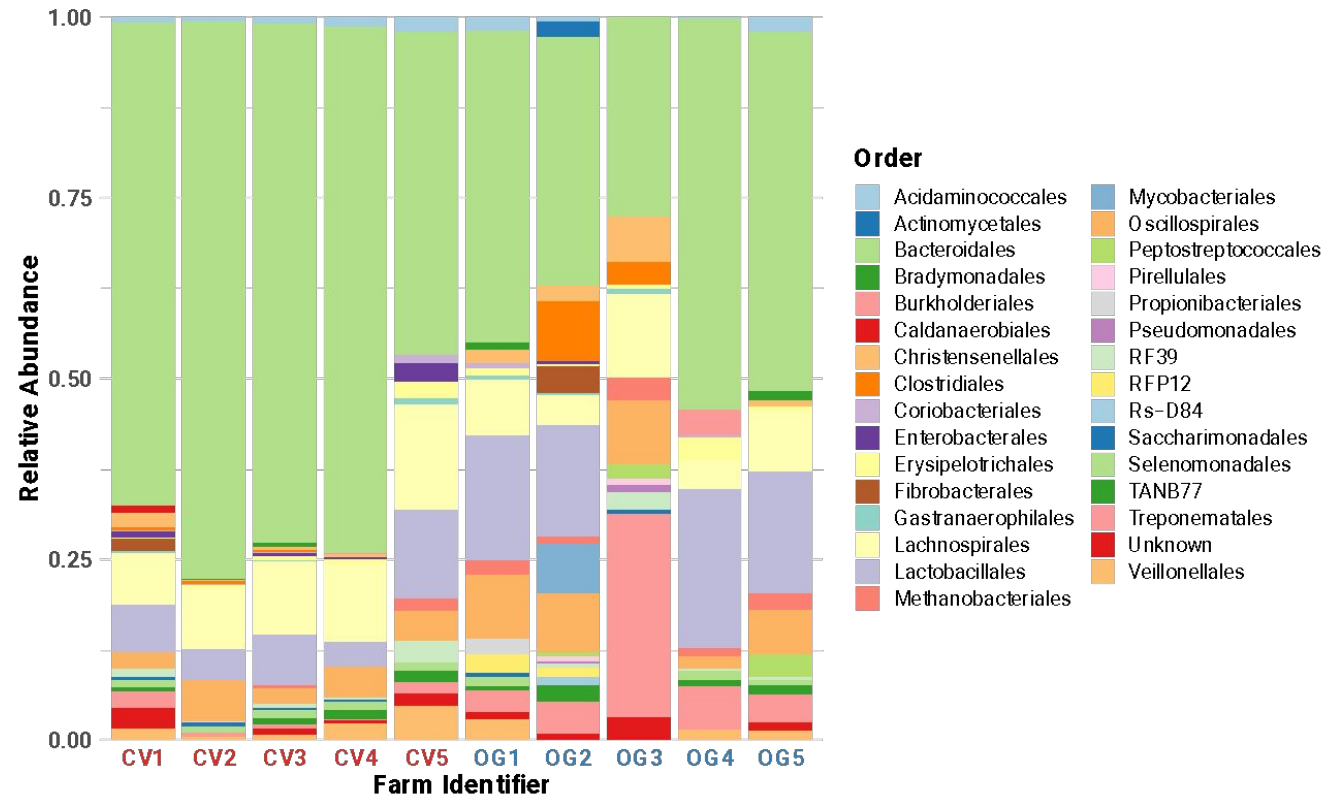
Step 2 - Sampling

- Plan your sampling well in advance
- Keep sampling procedure as consistent as possible
- “Freeze” the state of your community as fast as you can
 - Freezing (-80 °C)
 - Lyse cells
 - Stop the metabolism / growth / dying
- Always keep your needs in priority but try to save the sample for alternative approaches
- **Collect as much metadata as you can**
- While sampling is usually non-invasive you may need serious paperwork (HTA)

Metadata for the outlier



Metadata for the outlier



Step 2 - Sampling

- Plan your sampling well in advance
- **Collect the appropriate sample** (e.g., faecal microbiome vs. gut microbiome)
- Keep sampling procedure as consistent as possible
- “Freeze” the state of your community as fast as you can
 - Freezing (-80 °C)
 - Lyse cells
 - Stop the metabolism / growth / dying
- Always keep your needs in priority but try to save the sample for alternative approaches
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Sampling complex microbial communities



Sampling complex microbial communities



Step 3 – Laboratory procedures

- Get to the macromolecule phase as quick as you can
 - ‘Remove the environment’
- Degradation, growth, cell death, DNA fragmentation
- Contamination in low abundance / highly diluted samples
- Bacterial cells resist to cell lysis differently
- As soon as you have the DNA, further steps are the standard NGS library preparation and sequencing steps

Identifying the **placental** reagent “microbiome”

Identifying false signals in low microbial biomass studies.



A former representation of the placental microbiome



Over-calling fetal microbial exposure
Cell (2021)

Batch effects account for the main findings of an in utero human intestinal bacterial colonization study
Microbiome (2021)

Human placenta has no microbiome
but can contain potential pathogens
Nature (2019)

Recognizing the reagent microbiome
Nature Microbiology (2018)

The “fetal microbiome” and pitfalls of low-biomass microbial studies (2022)

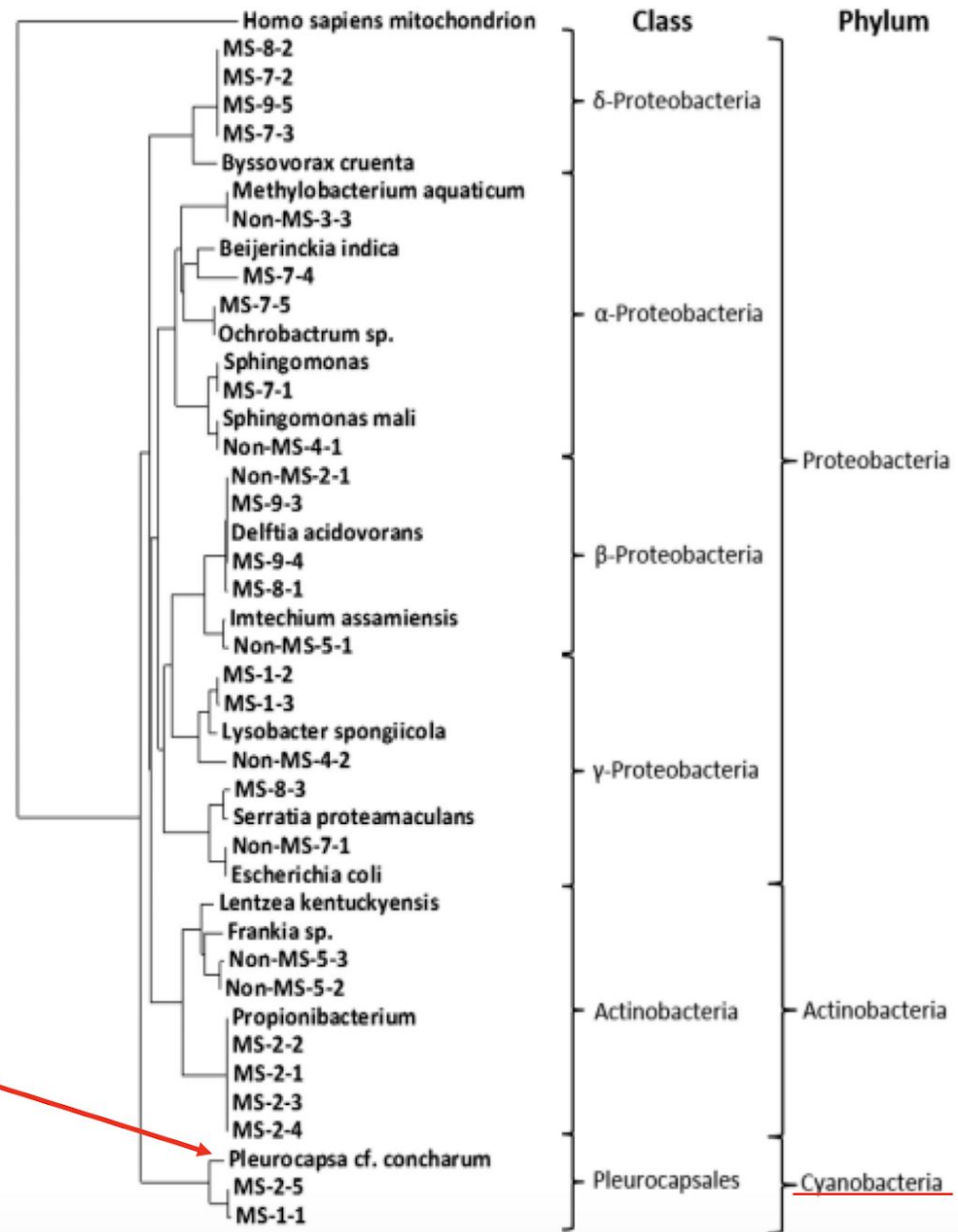
The Brain microbiome

SCIENTIFIC REPORTS

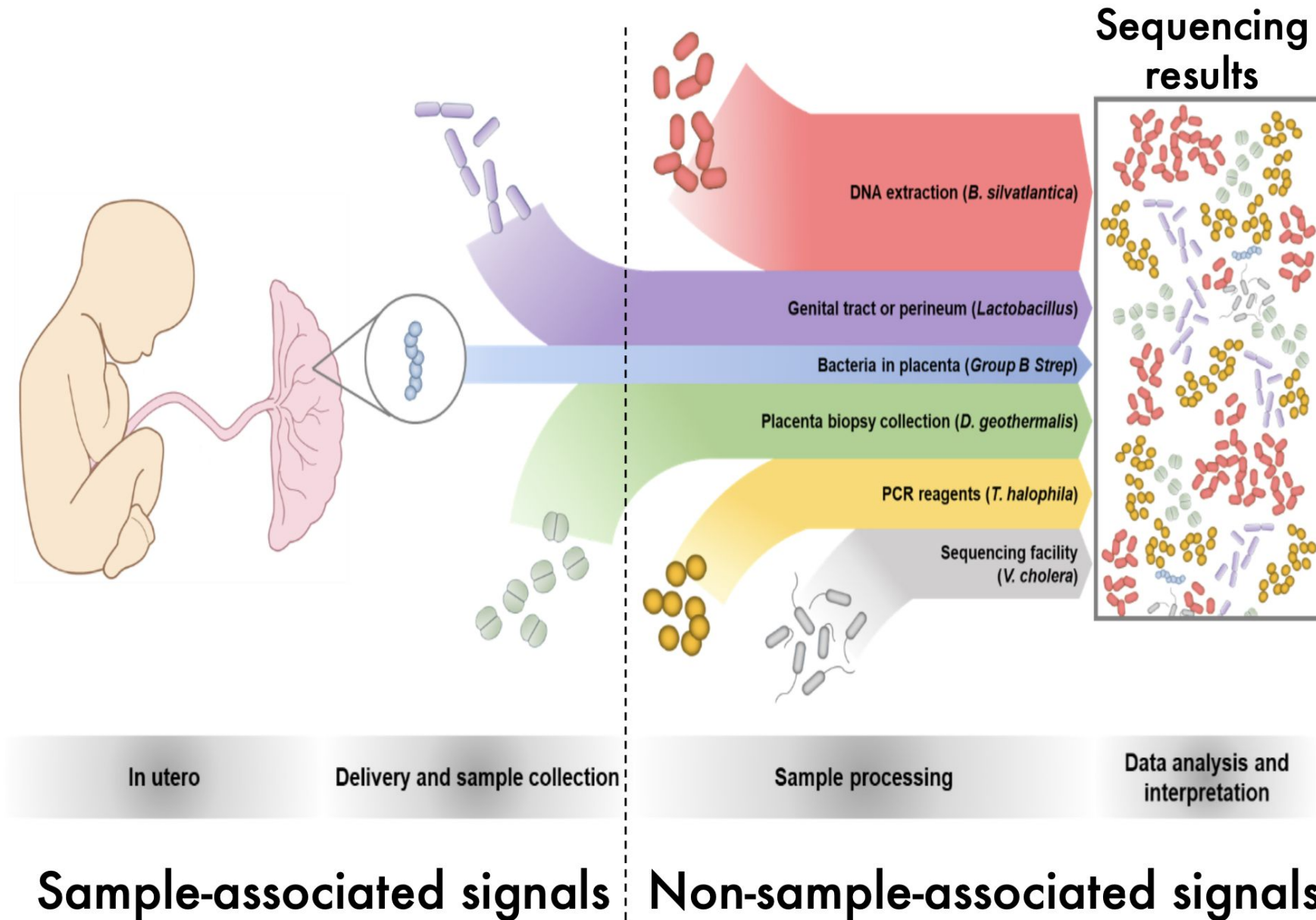
OPEN Brain microbiota disruption within inflammatory demyelinating lesions in multiple sclerosis

All the typical contaminants:
water, soil (environmental) & skin bacteria

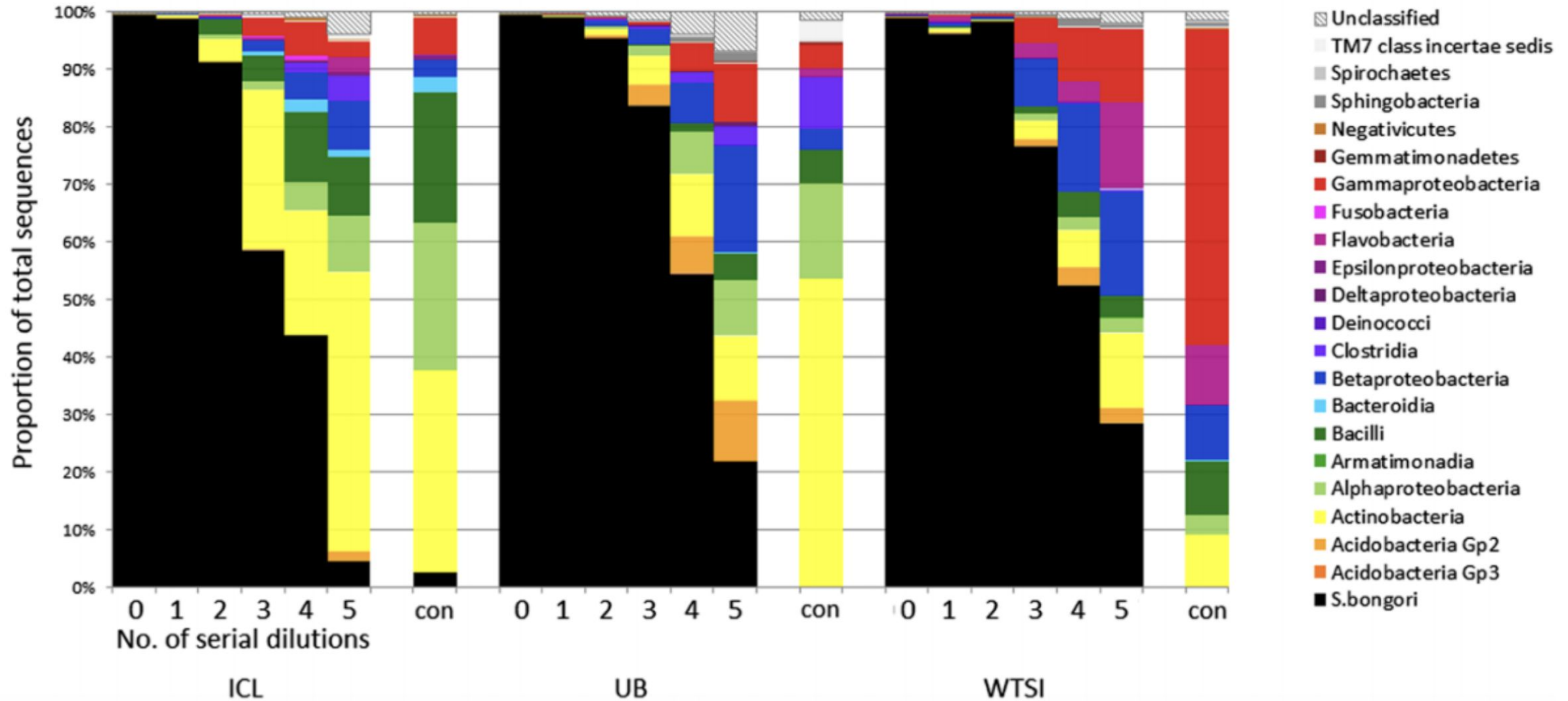
Also some photosynthetic Cyanobacteria:
(because people are so bright)



Sources of bacterial (contamination) signals



Reagent contamination 101, serial dilution.



Serial dilution of a *Salmonella bongori* spiked sample, analyzed in 3 separate labs.

Database contamination

- Carp sequences started to come up in nearly every type of sample when aligned to NCBI sequence databases
- It's not actually carp - they were adapter sequences

Article | [Open access](#) | [Published: 21 September 2014](#)

Genome sequence and genetic diversity of the common carp, *Cyprinus carpio*

[Peng Xu](#), [Xiaofeng Zhang](#), [Xumin Wang](#), [Jiongtang Li](#), [Guiming Liu](#), [Youyi Kuang](#), [Jian Xu](#), [Xianhu Zheng](#), [Lufeng Ren](#), [Guoliang Wang](#), [Yan Zhang](#), [Linhe Huo](#), [Zixia Zhao](#), [Dingchen Cao](#), [Cuiyun Lu](#), [Chao Li](#), [Yi Zhou](#), [Zhanjiang Liu](#), [Zhonghua Fan](#), [Guangle Shan](#), [Xingang Li](#), [Shuangxiu Wu](#), [Lipu Song](#), [Guangyuan Hou](#), ... [Xiaowen Sun](#)  [+ Show authors](#)

[Nature Genetics](#) **46**, 1212–1219 (2014) | [Cite this article](#)

Step 4 – Bioinformatics pipelines

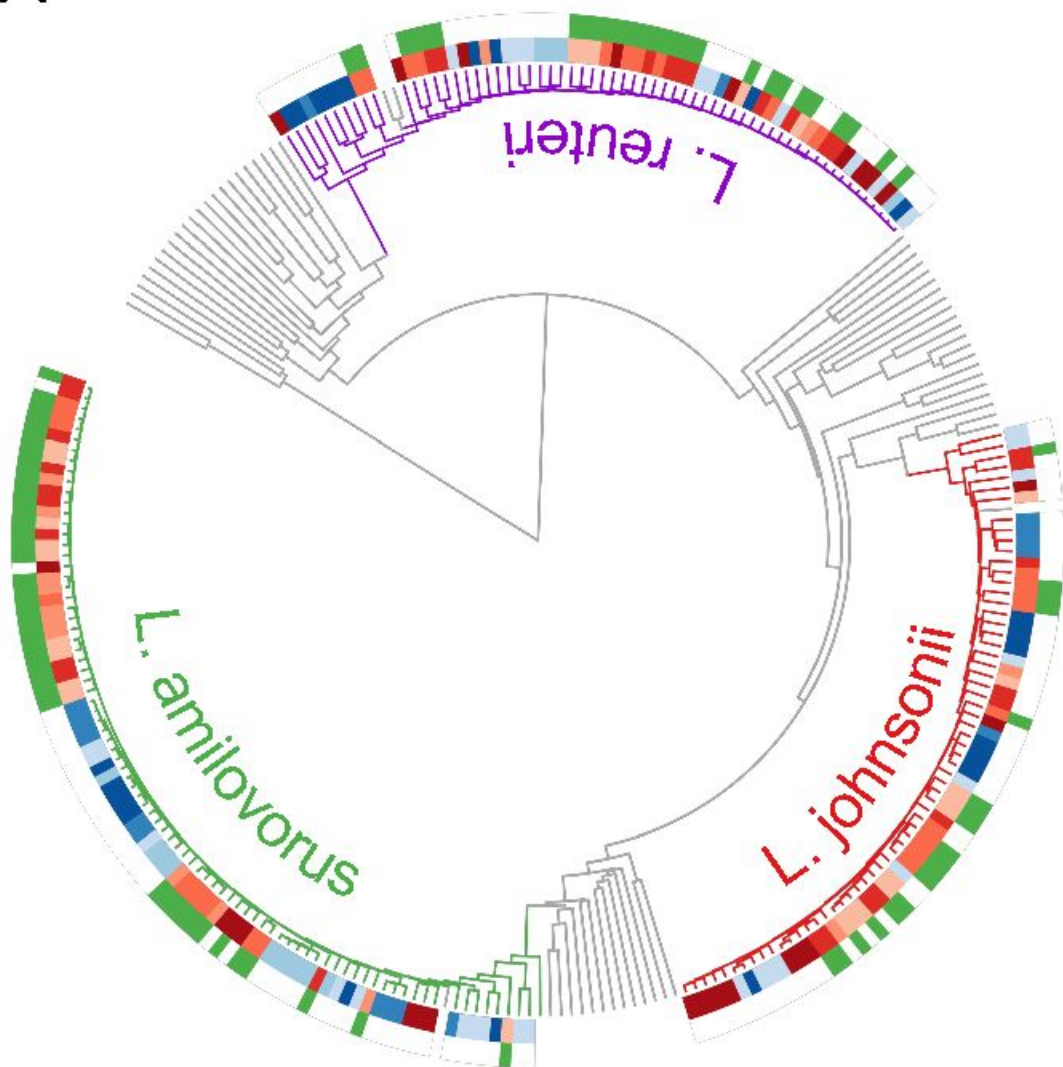
- From workstation to **high performance cluster** computing
- Tools usually require a lot of
 - CPU cores
 - RAM memory
 - Storage space
- Often need advanced bioinformatics knowledge
- Sometimes a dedicated bioinformatician
- Advanced visualisation skills (general and specific) and programming
- Tools, methods, applications evolve continuously
- Prepare for frustration on data analysis

Step 5 - Interpretation

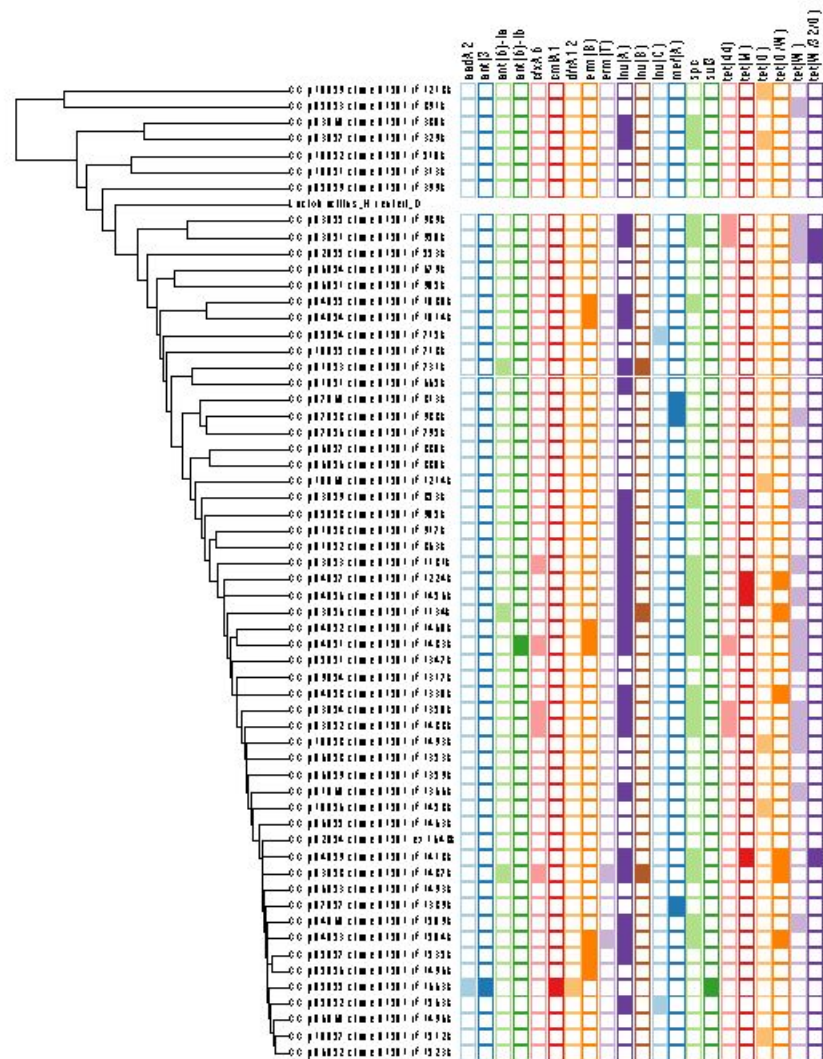
- Standard QC graphs and tables
- PCA / MDS plots, diversity measures, differential abundance analysis
- From a list of species to hundreds of new genome assemblies
- Taxonomic composition diagram
- Genetic distance tree (usually annotated)
- Project specific results, reports

Annotated trees

A



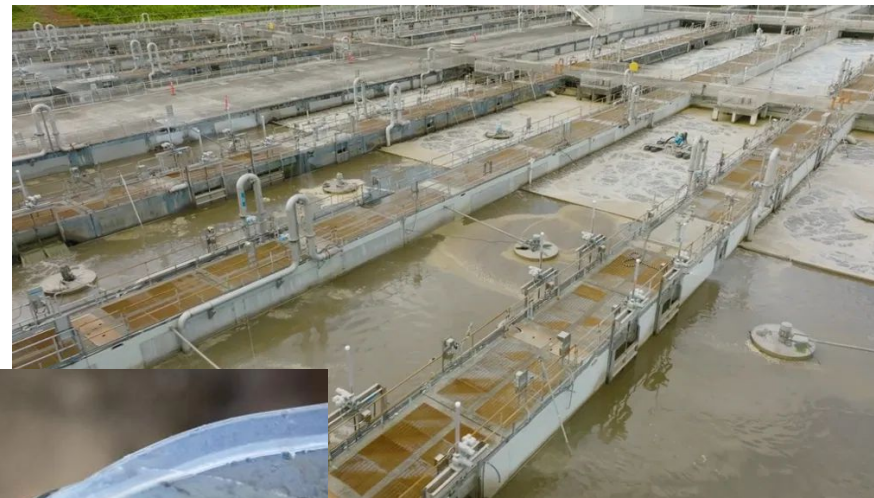
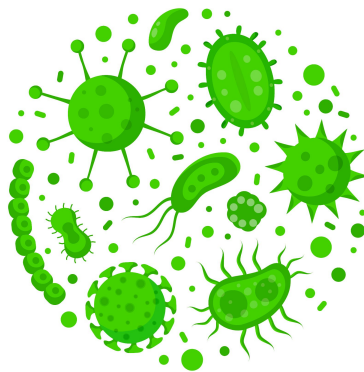
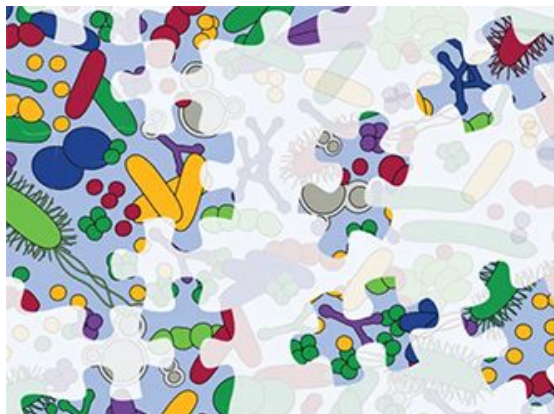
B



Cumulative bias in metagenomics

- Environmental bias
- Sampling and sample handling bias
- Extraction methods and protocols
- Sequencing platform specific bias
- Different algorithms, databases and approaches in bioinformatics
- High chance of bioinformatics software chains

Loss of information



Alternative / supplementary -omics

- Meta-metagenomics
- Metatranscriptomics
- Metaproteomics
- Meta-metabolomics
- Spatial metagenomics
- Culturomics